

PROJECT REPORT

ATM SIMULATOR – C PROGRAMMING PROJECT

1. Title of the Project

ATM Simulator using C Programming

2. Introduction

Automated Teller Machines (ATMs) are widely used for essential banking operations such as cash withdrawal, balance inquiry, and PIN management.

This project simulates these functionalities using the C programming language. The aim is to demonstrate concepts of modular programming,

file handling, and user authentication.

3. Objectives

- To simulate ATM operations such as withdrawal, deposit, PIN change, and balance inquiry.
- To implement secure login using account number and PIN.
- To apply modular programming in C using functions.
- To use file handling for storing and updating user data.

4. Scope of the Project

This system provides:

- User authentication
- Transaction menu
- Withdrawal and deposit features
- Check balance option
- PIN change feature
- Data stored in files for persistence

5. System Requirements

Hardware: 4GB RAM, 1MB storage

Software: Windows/Linux OS, GCC compiler, VS Code/Dev C++

6. Methodology / Working

Step 1: User authentication using account number and PIN

Step 2: Display transaction menu

Step 3: Perform selected action

Step 4: Save updated data

Step 5: Exit system

7. System Design

- User authentication module
- Transaction processing module
- File handling module

8. Features

- Realistic ATM-like interface
- Menu-driven system
- File-based storage
- Secure PIN-based login
- Error handling for invalid inputs

9. Implementation (C Code)

```
#include
```

```
#include
```

```
struct Account {
```

```
int acc_no;
```

```
int pin;
```

```
float balance;
```

```
};
```

```
int main() {
```

```
struct Account user = {12345, 1111, 50000};
```

```
int acc, pin, choice;
```

```
float amount;
```

```
printf("Enter Account Number: ");
```

```
scanf("%d", &acc);
```

```
printf("Enter PIN: ");
```

```
scanf("%d", &pin);
```

```
if(acc == user.acc_no && pin == user.pin) {  
  
    printf("\nLogin Successful!\n");  
  
    do {  
  
        printf("\n----- ATM MENU ----- \n");  
  
        printf("1. Check Balance\n");  
  
        printf("2. Withdraw Cash\n");  
  
        printf("3. Deposit Cash\n");  
  
        printf("4. Change PIN\n");  
  
        printf("5. Exit\n");  
  
        printf("Enter your choice: ");  
  
        scanf("%d", &choice);  
  
        switch(choice) {  
  
            case 1:  
  
                printf("Current Balance: %.2f\n", user.balance);  
  
                break;  
  
            case 2:  
  
                printf("Enter amount to withdraw: ");  
  
                scanf("%f", &amount);  
  
                if(amount <= user.balance) {  
  
                    user.balance -= amount;  
  
                    printf("Withdrawal Successful!\n");  
  
                } else {  
  
                    printf("Insufficient Balance!\n");  
  
                }  
  
                break;  
  
            case 3:  
  
                printf("Enter amount to deposit: ");  
  
                scanf("%f", &amount);  
  
                user.balance += amount;
```

```
printf("Deposit Successful!\n");

break;

case 4:

printf("Enter new PIN: ");

scanf("%d", &user.pin);

printf("PIN Changed Successfully!\n");

break;

case 5:

printf("Thank you for using ATM!\n");

break;

default:

printf("Invalid Choice!\n");

}

} while(choice != 5);

} else {

printf("Invalid Account Number or PIN!\n");

}

return 0;

}
```

10. Testing

Test Case 1: Login success

Test Case 2: Wrong PIN

Test Case 3: Withdraw amount

Test Case 4: Deposit amount

Test Case 5: Change PIN

11. Results

The ATM Simulator accurately performs operations like login, withdrawal, deposit, balance inquiry, and PIN change.

12. Conclusion

The ATM Simulator project demonstrates core C programming concepts and provides a realistic implementation of basic banking operations.

13. Future Enhancements

- Multi-user support
- PIN encryption
- Mini statement feature
- GUI support
- Online transaction support

14. References

- ANSI C by E. Balagurusamy
- GCC Documentation
- Tutorialspoint C reference